Introduction of Adeept Robotic Arm

With the rapid advancement of modern technology, robotic arms have been widely adopted in manufacturing, daily life, entertainment, teaching and scientific research, serving as a core application carrier in intelligent technology. Now let’s take an in-depth look at the core features and practical value of this robotic arm!

This robotic arm can not only execute intricate precision movements beyond human physical capability, but also repeat all types of tasks almost non-stop. It maintains stable and efficient operation continuously as long as regular maintenance is performed and power supply remains uninterrupted.

To help users get started quickly and master robotic arm operation effortlessly, we provide a comprehensive resource package filled with detailed graphic tutorials and complete program codes, guiding you step-by-step to master a full range of core skills as listed below:

- Get familiar with the Adeept robotic arm control board, fully learn the functions and usage of all its interfaces to lay a solid foundation for hardware operation;
- Learn to match the control board with various sensors through practical course cases to realize intelligent sensing and linkage of the robotic arm;
- Complete full assembly of the robotic arm from scratch following tutorials to build your own intelligent robotic manipulator;
- Master potentiometer control to precisely adjust the rotation angle and movement speed of the arm for basic manual operation;
- Grasp IR remote control operation to break free from wired connections and conveniently trigger start/stop and motion commands wirelessly;
- Utilize the ESP8266 module to establish communication between Arduino and Processing upper computer programs, enabling remote intelligent control of the robotic arm;
- Delve into the principle of Inverse Kinematics (IK), learn to convert spatial coordinates into servo angles, and operate the arm via mouse input to complete delicate tasks such as handwriting and drawing.

In addition, we have specially integrated an EEPROM hardware storage module onto the robotic arm control board to realize teaching and memory functions. Once programs with teaching & memory logic are flashed into the control board, the device can fully record every manipulation step and automatically replay the entire sequence of movements afterward, greatly boosting operational efficiency and playability.

This robotic arm balances practicality and fun perfectly. Whether used as a teaching aid to help learners understand mechanical control and programming principles, or as a creative tool for various interactive projects, it delivers an outstanding user experience.

We have fully optimized safety and usability during product development. Featuring a lightweight body paired with a stable base, the arm resists tipping during operation. The output torque of servos is finely calibrated, so even beginners can operate it safely without risk. All accessories pass strict quality inspection with standardized, user-friendly interfaces. Assembly and disassembly are intuitive, and routine maintenance is hassle-free. You can fully focus on operation and learning without worrying about equipment malfunctions.

Regarding the learning path, our tutorials are arranged in the following order. We recommend assembling the robotic arm in this sequence.

01 Adeept Robotic Arm Control Board	7/17/2026 11:48 AM	File folder
02 Robotic Arm Assembly	7/17/2026 11:41 AM	File folder
03 Robotic Arm Control	7/17/2026 11:49 AM	File folder

- Learn about the Adeept robotic- arm control board: Get a thorough understanding of the main control board by setting up the Arduino development environment, reading the control- board pin introduction, and running single- sensor examples.
- Robotic- arm assembly: Build the robotic arm step- by- step following the tutorial.
- Robotic- arm control: This section introduces several control methods, including infrared control, potentiometer control, Processing, mobile APP and more. It also covers multiple features such as position recording, to bring the full performance of the robotic arm into play.